For Instructors: How to Organize a Fox Hunt

Introduction

- This is a guide for instructors — it's not intended to be shown to participants.
- This covers how you can organize a fox hunt for you radio club, neighborhood, community, etc.
- The goal is for radio novices to come out of the fox hunt having had a good time and learned a bit about radio wave propagation.

Overview: A Fox Hunt Requires...

- 1 to 2 instructors
- 5 to 15 participants
- 90 minutes to 2 hours
- A “fox” transmitter
  - I recommend (and use) a $79 PicoFox
- Ideally one Ham radio per participants
  - No license required bc it’s receive only

Overview: Optional Equipment

- There are several optional pieces of equipment, too.
  - A directional antenna
  - An attenuator – very handy for when participants are very close to the fox!
- Snacks, including water!

Finding a Location

- You want to find a park that's big enough without being too big.
- It should ideally have a bit of diversity in the terrain:
  - Trees and/or other foliage
  - Some structures of different size and materials (sheds, etc.)
  - Fences, etc.
  - Some change in elevation

Other Qualities to Look For in a Park

- Accessible
- Public bathrooms, ideally with water fountains
- Phone signal and internet
- Big enough for any planned activities
- No conflicting events that day
- At least one available picnic table for meeting, setup, etc.

Hiding the Fox

- You want to hide it well, but not *too* well.
- I will often start by putting it in a paper bag (fast food, etc.)
  - Where I live, this is perfect urban camouflage.
- I'll then put the bag somewhere toward one of the edges of the park, opposite of the meeting spot.
- The goal is for it to take people a bit of searching and walking to find it, but NOT to frustrate them!

Running the Fox Hunt: The Beginning

- Show up early so you have enough time to hide the fox without anyone seeing you!
- Then hang out at the picnic table or whatever you've designated the meeting point to be.
    - Keep your phone handy in case people text you trying to find you!
- When you have a critical mass of participants – ideally no more than 10 minutes or so after the official start time – get started!
- Hand out HTs to participants that are pre-tuned to the fox's frequency.

Running the Fox Hunt: The Middle

- Have participants turn on their radios!
- Give them a moment to see if they can intuit how to (start) finding the fox! You can demonstrate it if they don't figure it out in a few minutes.
  - The technique is to hold the HT up to your body so your body acts as a shield that blocks transmissions from behind you.
  - You can then go where the fox tone gets louder.
- Walk along with participants as they search for the fox.
  - Don't be afraid to let them go in the wrong directional for a bit.
  - Move between groups of participants to answer basic questions.

Running the Fox Hunt: Easy Mistakes

- Participants will inevitably end up far from the fox but close to the source of a loud signal from it.
- As you will be walking by them – especially those who are far away from the fox – you will have the opportunity to explain what may be happening.
- One common problem is the signal reflecting off something like a chain link fence, etc.
- Explain this before the participant gets too frustrated and then help them find their own way back toward the fox!

More Fox Hunting Techniques: The Null Point

- A more advanced way to search is to hold the HT horizontally, so the top (point) of the antenna is pointed (ideally) toward the fox.
  - When demonstrating this, you can explain that the antenna receives the *least* effectively in that direction.
  - In radio terms, this is the “null point.”
- Once you know the general direction where the fox is, you use this technique and try to find the direction where the fox is *quietest*.
- This technique demonstrates how an antenna receives with different effectiveness in different directions!

More Fox Hunting Techniques: Signal Discrimination

- One problem, especially as you get close to the fox, is that you cannot distinguish exactly where the signal is strongest.
- The null point technique is one solution.
- Another is to shift the receiving frequency by one increment (10kHz)
  - This will diminish the strength of reception enough to better hear in what direction the fox is transmitting.
- This is obviously akin to using an attenuator, which is another option.

Running the Fox Hunt: The End

- You should be walking alongside the participants this whole time.
- Eventually, they should be congregating in the same general area.
- When people are right by the fox they have a lot of trouble finding it!
- If they get within 5 to 10 feet of the fox without finding it, you should give them hints or just tell them where it is.
- This demonstrates to them how hard it can be to find a “loud” signal!
- Once it’s found, you can go back to the picnic table etc. to debrief, have snacks, and potentially have the fox finder hide the fox for a second round.

Always End with an Evaluation

- It's important to have a strong ending to an event, rather than just having it peter out.
- No later than the official end time, have people come back to the picnic table for an evaluation of the event.
- First, ask participants what they *liked* about the fox hunt.
  - Make sure to visibly write down all the feedback!
- Next, ask participants what could be *improved* next time.
  - Again, write down all the feedback.
  - NEVER disagree or contradict people giving feedback, or you will stop receiving (honest) feedback!
- Thank everyone for their time and officially end the event on time.